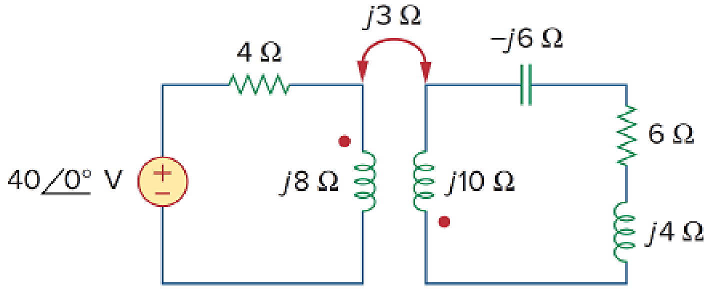
Problem

Find the input impedance of the circuit in Fig. 13.25 and the current from the voltage source.

Figure 13.25



Step-by-step solution

Step 1 of 3

Figure 1

Step 2 of 3

Write the expression for impedance $\mathbf{Z}_{in}$ seen by the voltage source.

$$\mathbf{Z}_{in} = \frac{\mathbf{V}_1}{\mathbf{I}_1}$$

$$= R_1 + j\omega L + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$$

$$= R_1 + j\omega L + \frac{X_M^2}{R_2 + j\omega L_2 + Z_L}$$

$$\mathbf{Z}_{in} = 4 + j8 + \frac{3^2}{j10 - j6 + 6 + j4}$$

$$= 4 + j8 + \frac{9}{6 + j8}$$

$$= 4 + j8 + 0.54 - j0.72$$

$$= 4.57 + j7.28$$

$$\mathbf{Z}_{in} = 8.58\angle 58.05^\circ\ \Omega$$

So input impedance $\mathbf{Z}_{in}$ is $\boxed{8.58\angle 58.05^\circ\ \Omega}$

Step 3 of 3

Calculate the value of the current $\mathbf{I}_1$.

Write the expression for input impedance $\mathbf{Z}_{in}$.

$$\mathbf{Z}_{in} = \frac{\mathbf{V}_1}{\mathbf{I}_1}$$

$$8.58\angle 58.05^\circ = \frac{40\angle 0^\circ}{\mathbf{I}_1}$$

$$\mathbf{I}_1 = \frac{40\angle 0^\circ}{8.58\angle 58.05^\circ}$$
$$\mathbf{I}_1 = 4.602\angle -58.05^\circ$$

The current $\mathbf{I}_1$ from the voltage source is $\boxed{4.602\angle -58.05^\circ \text{ A}}$.